

Plexus Discovery

Progress note — written to be read, not decoded

31 July 2026

Plain English throughout. Organised by phase, because phases are where we stop and you decide.
The technical specification is preserved separately in `plexus2_technical.pdf`.

1. What we are doing — two things at once

Track A — build the method. An automated loop that searches for biological *mechanisms* rather than parameter values. It proposes a change to the model’s structure (“remove the pushing force”, “add oriented cell division”), writes down what it expects *before* running, runs the simulation, measures, and records whether it was right. The discipline that makes it worth building: a change of numbers can never masquerade as a new idea.

Track B — reproduce Okuda’s result, as the proof that Track A works. Okuda et al. (2018) grew a hollow ball of cells that spontaneously forms tubes, undulations and branches, driven by a chemical pattern on its own surface. We are rebuilding that here. If the loop finds the mechanism largely by itself, the method is demonstrated. *The figure is not a side quest — it is the evidence.*

The two tracks feed each other: the reproduction is the honest test of the method, and the method is what makes the reproduction more than hand-tuning.

2. Where we actually are

Track A is further along than Track B, and until this morning I was over-reporting both.

An independent review this morning found that **three of the measurements the loop relies on are broken**, and you have since found a fourth. Not the simulations — the *rulers*. A large part of what I reported over the last two days was measured with faulty instruments and does not currently stand up.

The clearest example, and why you were right to stop me.

I ran 32 simulations overnight, sweeping five settings, and reported a strong conclusion: *“making the chemistry shape the tissue and keeping the tissue intact are mutually exclusive.”* Every one of those 32 runs ended with exactly **1778 cells**. I noticed that, wrote it down, and called it “remarkable”. I never asked the obvious question: *why is it identical every time?*

The answer is arithmetic. The memory buffer for the mesh holds 3552 points, and the geometry of a closed cell sheet fixes the largest possible number of cells at $(3552 + 4)/2 = 1778$. **Every run hit the wall and stopped.** I was not measuring biology. I was measuring the size of an array.

That conclusion is now **withdrawn**. It may still turn out to be true — but it is not measured, and I presented it as if it were.

3. What we believe about the biology

Nothing. The register is empty.

Every measurement we have was taken with at least one of the four broken rulers in the path. There is no scientific claim in this document, and there should not be one until Phase 1 has re-measured with instruments that work.

An earlier draft of this section had a table grading five claims as “solid”, “weak” or “unsupported”. Cedric was right to strike it, and the reason is worth recording.

Grading claims on a confidence scale is the correct move when the instruments are sound and the evidence is thin. It is the *wrong* move when the instruments themselves are broken — then the register should be *voided*, not annotated. Grading felt like rigour, but it smuggled the conclusions forward: a reader skims “solid”, carries it, and the broken ruler is forgotten two pages later. If we cannot say how wrong a number is, it does not get a grade; it gets deleted.

What we do know — about our own code, not about biology

These are facts established by *reading the source*, not by measuring anything. They are a different kind of statement, and they belong in a different list.

- **In the two front-page movies, the chemistry cannot affect the shape.** The switch that lets the chemical drive growth is set to 50, and the chemical is a quantity of order 1. The switch term is therefore about 10^{-9} — off for any value the chemical could take. Separately, the growth setting is at its uniform limit, so every cell grows identically regardless. Both are visible in the configuration file; neither needed an experiment.
- **We were never running Okuda’s system (§7).**
- **The mesh buffer is sized as a multiple of the starting cell count**, so a run that begins with 150 cells can never exceed 1778. That is why all 32 runs stopped there.

None of these tells us anything about tubes. They tell us the experiment we thought we were running is not the experiment that ran.

4. The four broken rulers

All four are *measurement* faults. The physics may be fine; we simply could not see it. The fourth is the camera, and you spotted it by looking at the movies — it is the ruler the automatic eye-check reads, and it is broken in two independent ways.

1. **The “does the shape survive?” test was measuring a fresh sphere.** The intent: grow the tube, switch off growth and pushing, let it relax, see what survives. A real tube survives; an artificially forced one collapses. The code was meant to continue from the end of the run — instead it started a brand-new simulation from the initial sphere, and so measured the starting condition every time. It returned essentially the same value (1.014) in 14 of 16 runs. **That value carries full weight in the score that ranks every experiment.** *Not yet fixed — one of Phase 0’s open items.*
2. **Two different quantities were both called “how far it sticks out”.** One measured distance from the centre of *the tissue*; the other from the *origin of the coordinate system*. The ball drifts sideways as it grows, so the second reads drift as elongation. *Fixed this morning.*
3. **A safety check existed, but not where the danger was.** We once chased a phantom result for days, caused by pairing mismatched frames. A guard was written to catch it. My overnight study used exactly the pattern that guard forbids, in three places, and never called the guard. *Fixed this morning.*
4. **The camera hides the thing we are looking for, and hides growth entirely.** (*Your observation, confirmed in the code.*) Two separate faults, both in the movie every automatic eye-check reads:
 - **One fixed viewpoint.** The campaign’s movies are drawn from a single angle that never moves. A tube growing away from that angle sits *behind the body* and is simply not in the picture. There is no second view — no top-down — to catch it.
 - **The zoom follows the object.** The frame is re-fitted to the tissue on *every frame*, so a ball that doubles in size is drawn the same size throughout. **Growth is invisible by construction.** Nothing that watches these movies — human or machine — can see the thing the whole campaign is about.

Okuda’s figures use a fixed frame and a consistent viewpoint, which is exactly why they read clearly. *Not fixed — now part of Phase 0, because it is an instrument, not a nicety.*

5. What is genuinely working

The corrections should not hide the parts that earned their keep.

- **Predictions are written down before runs, and wrong ones are recorded as wrong.** Twice in two days I proposed a mechanism and the measurement killed it within the hour. Both sit in the record as retractions rather than being quietly edited away. This is the part I would defend hardest.
- **The video check found a disagreement — but I over-claimed who was right.** On the first full round the system described each movie in words automatically and compared that to the numbers. It *vetoed the two highest-scoring runs*, including one that three separate analyses had called a tube. I reported this as “the numbers were wrong and the picture was right”. **That was too strong.** Given ruler 4 above, the picture may have been hiding a tube behind the body, or failing to show growth at all. What the check genuinely established is that *the two*

instruments disagree — which is still valuable, and is precisely why it exists. Which one to believe is not yet decided, and cannot be until the camera is fixed.

- **The loop refuses to guess.** Asked to score 32 results against predictions written in a unit it did not recognise, it declined 32 times rather than inventing 32 confirmations.

6. The team of agents — what is actually switched on

You were told 6 of 12 are implemented or used. I audited it directly against the code rather than accept the figure, and it is better than that — but there are two real gaps, and they are worth closing exactly as you say.

The design calls for 15 roles (14 automatic, plus a human who patches the substrate from outside the loop). **All 15 are written. 13 actually run.** The two that do not:

Role	What it is for	Status
Evolution	Take the round's winner and improve it — simplify it, combine it with a runner-up, ground it in the paper. Without it the loop can only ever <i>pick</i> from what it happened to propose; it can never <i>refine</i> .	Written, never called anywhere .
Referee	Rank a batch by a proper tournament (every candidate against every other, aggregated) rather than by sorting on one number. The tournament is what protects against ordering bias.	Written, but only ever exercised in its own self-test . The live round sorts on a single score instead.

The other thirteen — the one that holds the objective, the one that reads the papers, the one that proposes, the peer-reviewer, the type-checker, the second-opinion judge, the duplicate detector, the three independent analysts, the eye-check, the one that writes the causal story, the pattern-distiller, and the one that certifies the instruments — all ran on the last round.

Two honest caveats: the eye-check is currently reading a broken camera (ruler 4), and the time budget is not merely unenforced — it is not even measured.

How long the agents take — nobody knows, and the arithmetic is alarming

You asked whether the time per call is reasonable, and suggested five minutes a round as the target. Checking it produced a short answer and a long one.

The short answer: we do not measure it. The last round ended by reporting `calls: 0, minutes: 0.0` — for a round that made about twenty-five calls. A time ledger exists, and no agent is routed through it. So the declared ceiling of 25 minutes has never constrained anything, and there is no record of how long any call has ever taken.

The long answer: the declared allowances add up to nearly two hours. These are timeouts rather than expected durations, but they are what the system would permit before intervening, on a round with five scored runs:

Role	Allowance	Calls per round	Worst case
Reads the papers	4 min	1	4 min
Proposes the batch	10 min	1	10 min
Peer-reviews it	5 min	1	5 min
Analysts	4 min	15 (3 per run)	60 min
Eye-check	3 min	5	15 min
Second opinion	2 min	1	2 min
Writes the story	6 min	2	12 min
Distils the patterns	8 min	1	8 min
		27 calls	116 min

Why five minutes is nonetheless achievable. Twenty of the twenty-seven calls are the per-run ones — three analysts and an eye-check, for each of five runs — and they are completely independent of one another. They run one after another today for no reason other than that the code was written as a loop. Run them together and that whole block costs the time of its slowest single call rather than the sum of twenty.

Adding the change already on the list — starting the next round’s proposal *while* the current simulations are still running, so its ten minutes leave the critical path entirely — a round’s unavoidable serial cost is roughly: peer-review, the slowest analyst, the second opinion, two stories in parallel, and the distillation. **About four to five minutes, which is your number.**

Two items follow, and they are separated deliberately: *measuring* is cheap and goes into Phase 0 with the other instruments; *making it fast* is real work and goes into Phase 2b.

7. What stands between us and the Okuda figure

I previously framed this as an exotic parameter-range problem. **Cedric’s reading is closer to right: most of the gap is mundane, and it is about the starting conditions.** That matters, because mundane gaps get closed in a day.

The mundane part — and it is most of it

Starting condition	Okuda	What we ran	Fix
Number of cells at the start	200, 2000, 4000	150	set the number
Room to grow into	grows from there	capped at 1778	bigger buffer
Number of spots at the start	~5 at 2000 cells	never counted	count them
Size of those spots	set by his figures	never measured	measure them

We were not running his system. The overnight study began with 150 cells — his smallest case starts with 200 and his undulation case with 2000. And because the mesh buffer is sized as a multiple of the *starting* count, beginning at 150 capped us at 1778 cells. His 4000-cell case is not reachable at all today. None of this is subtle physics; it is four numbers.

The same applies to the pattern. Rather than trying to copy his length-scale setting — which in our code is not even the same quantity as his — **we match what he reports seeing:** about five spots on a 2000-cell ball. Tune our diffusion until we get five spots at 2000 cells, then freeze it. That is a calibration against an observation, and it sidesteps the units problem entirely.

What is left after that

- **Two ranges are simply too narrow** — the sharpness of the growth switch (he uses 10, we allow up to 8) and how far the inhibitor spreads (he uses 10, we allow up to 2). Widening two numbers.
- **One real gap.** In Okuda’s model the chemistry responds to the tissue’s *shape*: chemical flows between neighbouring cells in proportion to how much wall they share. Ours ignores geometry entirely, so patterning cannot react to deformation. This probably does not block the plain tube, but it is likely to matter for *branching*, where domains split as cells stretch. This is the one item that is genuine work rather than a number change.

8. The phases

Eight stages. **I stop at the end of each one and wait for you.** That is the partition.

Phase 0 — Pre-flight: fix the instruments

IN PROGRESS (2 of 6)

Goal. Make the measurements trustworthy. Nothing runs unattended until this lands.

Why. Running before this writes uninterpretable results into a permanent record. Four rulers were wrong; two are now right.

Remaining.

- **The “does the shape survive?” test** (§4, item 1). Either repair it so it continues from the end state, or remove it from the score. A constant carrying full weight is worse than no term at all.
- **The camera** (§4, item 4). Fix the frame so it does not follow the object — one size, held for the whole run and shared across runs, as in Okuda’s figures — and add a second viewpoint so a tube cannot hide behind the body. This was originally scheduled much later as a presentation nicety. It is not: it is the ruler the eye-check reads, and until it is fixed we cannot say whether the eye-check’s vetoes were right.
- **Start from Okuda’s cell counts** (§7). Set the starting number to his, and size the buffer for where the run is going rather than where it began. Four numbers.
- **Put a stopwatch on every agent call** (§6). Route them through the existing time ledger so the next round reports what it actually cost. Measuring only — speeding it up is Phase 2b.

What you get. A one-line confirmation that the rulers are sound, plus one before/after movie so you can see the camera change with your own eyes.

Phase 1 — Re-measure what Phase 0 invalidated

NOT STARTED (~1 hour)

Goal. Re-run the 32-simulation study with a buffer ten times larger and the frame-pairing guard live.

Why. This is the phase that decides whether my overnight conclusion was real.

What it will decide. One of two outcomes, and *both are good*:

- it survives — we have a genuine structural result, which is exactly what a paper about the method wants; or
- it dissolves — and we avoided publishing an artefact.

What you get. The answer in a sentence, plus the corrected table.

Phase 2 — Make Okuda's settings reachable

NOT STARTED (~1 week)

Goal. Widen the three out-of-range settings; replace the mislabelled length-scale with the quantity Okuda actually varies; add the missing shape-to-chemistry feedback; add a fixture that asserts his published constants rather than tuning them.

Why. Without this, Track B cannot succeed at any effort level.

What you get. A list of what now matches his paper and what still does not, with reasons.

Phase 2b — Switch on the two agents that never run

NOT STARTED (1 day)

Goal. Put **Evolution** and the **Referee** into the live round (§6); run the twenty per-run calls together instead of one after another; move the proposal off the critical path by starting it while the simulations run. Target: **a round costs five minutes of agent time**, measured, not asserted.

Why. Without Evolution the loop can only pick among the changes it happened to think of, never improve the best one — which is half of what makes it a search rather than a lottery. Without the Referee's tournament, ranking rests on a single number, and that number is the one Phase 0 is currently repairing.

What you get. Confirmation that all fourteen automatic roles ran in one round, and the measured wall-clock against the five-minute target.

Phase 3 — Measurements that can tell his regimes apart

NOT STARTED (2 days)

Goal. Four measurements: how many spots, how thick the tube, how many branches over time, and the shape of the surface. Calibrate against his roughly five spots at 2000 cells, then freeze the calibration.

Why. We currently have no measurement that can distinguish a tube from an undulation from a branch — precisely the three outcomes the figure has to show.

What you get. The four numbers, checked against his published figures.

Phase 3b — Point the camera *along the tube*

NOT STARTED (half a day)

Goal. Phase 0 makes the camera honest — fixed frame, second viewpoint. This phase makes it *smart*: lock it to the tube's own axis, one view side-on and one looking straight down the tube.

Why. The end-on view *is* the thickness and branch-count measurement, so camera and measurement share one axis. It is a refinement, not a repair — the repair is in Phase 0.

What you get. Movies in which the tube is always in full profile, whichever way it grew.

Phase 4 — The demonstration

NOT STARTED (one wave, ~1 hour)

Goal. Run the designed grid — six to eight simulations — that produces the figure: tube, undulation and branching, at his settings.

What you get. *The figure.*

Phase 5 — The method claim

NOT STARTED

Goal. Write up what the loop found by itself versus what we told it, with the record of predictions and retractions as the evidence.

What you get. The argument that Track A works, resting on Track B.

9. What I need from you

Nothing blocking. Two things worth knowing:

1. **Phase 1 is the interesting one** and it costs about an hour of compute. When it finishes I will bring you the answer, not the reasoning — unless you want the reasoning, in which case say so and it becomes the default.
2. **Tell me if this note is the right shape.** Too long, too short, wrong order — it is cheap to change, and it is the thing that lets you steer.

Appendix — words I could not avoid

Only these four. Everything else above is ordinary English.

- **Cell sheet (or mesh)** — the model of the tissue: a hollow ball of polygonal cells sharing walls, like a football.
- **Operator** — one mechanism as a reusable building block: “cells divide”, “chemical spreads”, “surface pulls tight”. A model is a combination of these.
- **The loop** — the automated cycle: propose a change, predict the outcome, run, measure, record whether the prediction held.
- **Phase** — one of the eight stages above. Our agreed stopping points.

Technical detail — the control law, the agent roles, the operator language, the validation ladder and the full external review — is preserved in `plexus2_technical.pdf` (23 pp). My working log is `SESSION_LOG.md`. You should not need either.